REFINEMENT OF FPQC COVERINGS MAKES THE PULLBACK FUNCTOR ON DESCENT DATA FULLY FAITHFUL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 2 navigation stubs, for the single-covering refinement lemma and for the commutativity of the pullback family; the reduction that uses them is written out in the span.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The canonical tag of this label does not occur among the tags of the published items of descent.tex, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- Fully faithfulness of the pullback functor along a refinement of fpqc coverings. Same chapter, Lemma lemma-fpqc-refinement-coverings-fully-faithful, invoked at proof lines 9106-9107 and 9112-9114. Supplies the single-covering case to which the proof reduces.
- Commutativity of the pullback functors on descent data along a refinement. Same chapter, Lemma lemma-pullback-family part (3), invoked at proof line 9108. Supplies the two-commutative diagram relating the three categories of descent data.
- The category of descent data relative to an fpqc covering. Same chapter, the definition of descent data, named in the statement at lines 9083-9087. Supplies the categories whose functors are compared.

Source reference. Original lemma 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *Let S be a scheme. Let $\mathcal{U} = \{U_i \to S\}_{i \in I}$, and $\mathcal{V} = \{V_j \to S\}_{j \in J}$, be fpqc-coverings of S. If $\mathcal{U}$ is a refinement of $\mathcal{V}$, then the pullback functor*

$$\textit{descent data relative to } \mathcal{V} \longrightarrow \textit{descent data relative to } \mathcal{U}$$

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is fully faithful. In particular, the category of schemes over S is identified with a full subcategory of the category of descent data relative to any fpqc-covering of S.

Proof. Consider the fpqc-covering $\mathcal{W} = \{U_i \times_S V_j \to S\}_{(i,j)\in I\times J}$ of S. It is a refinement of both $\mathcal{U}$ and $\mathcal{V}$. Hence we have a 2-commutative diagram of functors and categories

$$\begin{array}{ccc} DD(\mathcal{V}) & \longrightarrow & DD(\mathcal{U}) \\ & \searrow \quad \swarrow & \\ & DD(\mathcal{W}) & \end{array}$$

Notation as in the proof of Lemma 0 and commutativity by Lemma 0 part (3). Hence clearly it suffices to prove the functors $DD(\mathcal{V}) \to DD(\mathcal{W})$ and $DD(\mathcal{U}) \to DD(\mathcal{W})$ are fully faithful. This follows from Lemma 0 as desired. □